OFFICIAL TECHNICAL SPECIFICATION · APEX SQUAD

NATIONAL 181 / 112 CAD INTEGRATION
Decision-Support Platform v2.4

SAATHI (साथी) — System Architecture & Tech Stack

10-Second Helpline Triage & Coordinated Multi-Agency Dispatch in Hinglish

ENGINEERING TEAM	PRIMARY DOMAIN	VERIFICATION	TRIAGE LATENCY
Apex Squad	Social Impact · Emergency CAD	21 / 21 PASS (100%)	< 1.2s Local / < 10s Total

1. Executive Summary & Core Innovations

SAATHI is a real-time computer-aided dispatch (CAD) decision-support workstation built specifically for operators of India's **181 Women Helpline** and **112 Emergency Response Support System (ERSS)**. Most traditional emergency AI tries to understand what a caller articulates explicitly. *SAATHI is engineered for the caller who cannot say it.*

INNOVATION 1 — Coded-Call Detection

Under intimate partner confinement, callers pretend to order food (*"bhaiya paneer roll bhej do"*) or call a relative. SAATHI extracts hidden distress indicators, assigns **P2 Urgent** priority, and restricts operator guidance to safe **Yes/No questions only**.

INNOVATION 2 — Invariant Urgency Floors

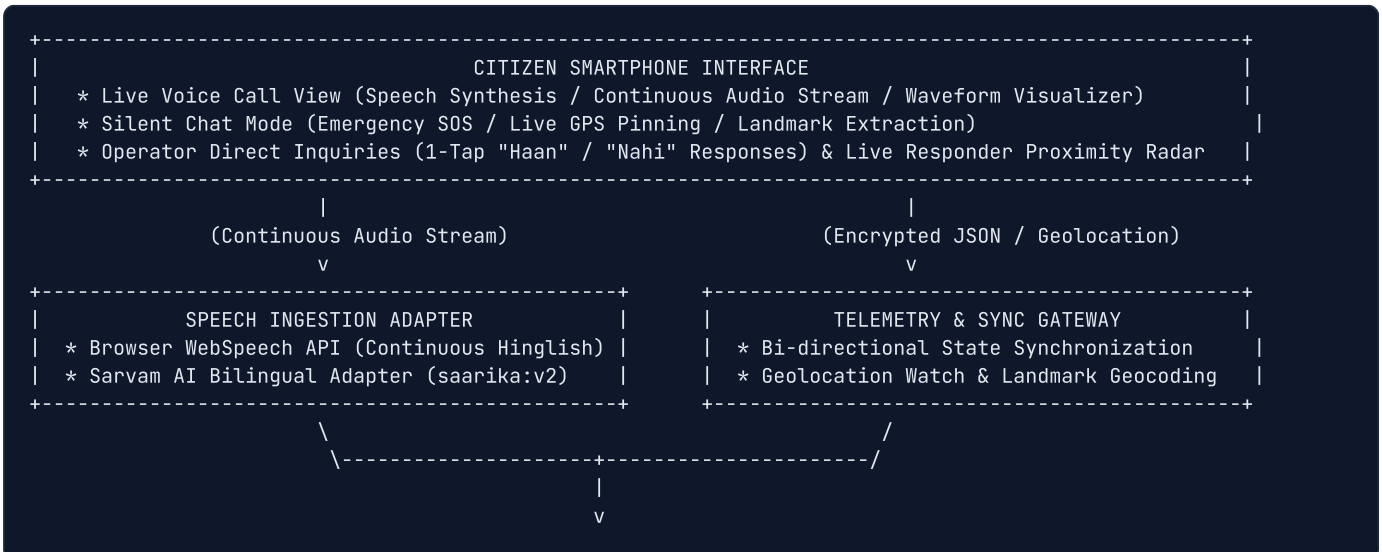
Once hard evidence of danger (weapons, active physical violence, life threat) is registered, the priority rating (e.g. **P1 Immediate**) is locked monotonically. If the caller subsequently claims *"sab theek hai"* under duress, the urgency floor **cannot be lowered**.

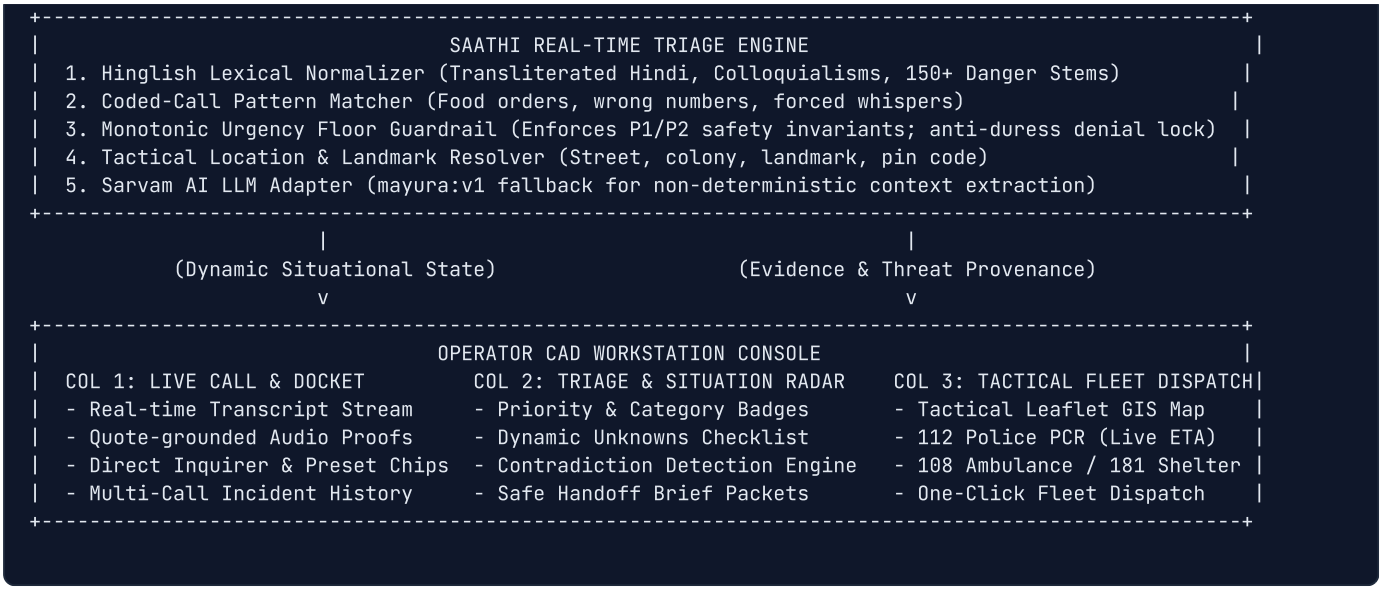
INNOVATION 3 — SafeSignal Protocol

Strict rule filtering (R1–R10) prevents questions that endanger callers. Prohibits abuser names, commands to move, promises, or multi-part questions. Supports 1-touch citizen responses (**Haan / Nahi**) transmitted directly to the caller's mobile interface.

2. End-to-End System Topology & Architecture

The system operates as an integrated monorepo connecting the citizen's mobile client, speech and language processors, deterministic invariant rule guards, and the operator's tactical command center.




SAATHI REAL-TIME TRIAGE ENGINE
1. Hinglish Lexical Normalizer (Transliterated Hindi, Colloquialisms, 150+ Danger Stems)
2. Coded-Call Pattern Matcher (Food orders, wrong numbers, forced whispers)
3. Monotonic Urgency Floor Guardrail (Enforces P1/P2 safety invariants; anti-duress denial lock)
4. Tactical Location & Landmark Resolver (Street, colony, landmark, pin code)
5. Sarvam AI LLM Adapter (mayura:v1 fallback for non-deterministic context extraction)
(Dynamic Situational State)
(Evidence & Threat Provenance)
OPERATOR CAD WORKSTATION CONSOLE
COL 1: LIVE CALL & DOCKET
- Real-time Transcript Stream
- Quote-grounded Audio Proofs
- Direct Inquirer & Preset Chips
- Multi-Call Incident History
COL 2: TRIAGE & SITUATION RADAR
- Priority & Category Badges
- Dynamic Unknowns Checklist
- Contradiction Detection Engine
- Safe Handoff Brief Packets
COL 3: TACTICAL FLEET DISPATCH
- Tactical Leaflet GIS Map
- 112 Police PCR (Live ETA)
- 108 Ambulance / 181 Shelter
- One-Click Fleet Dispatch

3. Technology Stack & Infrastructure Breakdown

LAYER	TECHNOLOGY / LIBRARY	ROLE & ARCHITECTURAL RATIONALE
Frontend Core	`React 19` + `TypeScript`	Strictly typed functional components with zero runtime overhead. Single-state reactive store syncing caller and CAD consoles.
Build Engine	`Vite 7.3.1`	Sub-second Hot Module Replacement (HMR) and optimized rollup bundle generation (~480 KB production footprint).
Tactical GIS & Maps	`Leaflet 1.9.4` + OpenStreetMap	Offline-capable vector mapping, caller GPS pin visualization, tactical radius rings (500m/1.5km), and fleet vehicle markers.
Icons & Telemetry	`Lucide-React`	Crisp, high-contrast tactical symbols for emergency dispatch (Siren, Radio, MapPin, ShieldAlert, PhoneCall).
ASR Ingestion	`Web Speech API` + Sarvam	Browser-native low-latency continuous recognition for Hindi and English, with Sarvam AI `saarika:v2` bilingual streaming adapter.
Natural Language (LLM)	Sarvam AI `mayura:v1` / `sarvam-2b`	Indian-context LLM fallback for ambiguous Hinglish syntax; structured JSON output extraction for case summaries.
Deterministic Engine	Custom Lexicon & Invariant State Machine	150+ colloquial Hinglish danger stems, sub-10ms deterministic execution, zero-internet offline reliability.
Testing Framework	`Vitest 3.0.0` + JSDOM	Automated verification of safety floors, coded-call detection, question filtering rules (R1–R10), and location parsing.

4. Component Monorepo Directory & Module Map

Client & CAD Components (`src/components/`)

- `OperatorCadView.tsx` : 3-column tactical workstation. Real-time transcript, dynamic unknowns checklist, custom question transmitter, and multi-agency fleet dispatch.
- `UserPhoneView.tsx` : Mobile smartphone container. Features realistic bezel, voice call visualizer, silent chat, live location sharing, and responder ETA radar.
- `TacticalMap.tsx` : Leaflet GIS map with dynamic markers for caller location and real-time positioning of 112, 108, 181, and DLSA units.

Core Engine & Services (`src/services/` & `data/`)

- `triageEngine.ts` : Rule engine executing urgency classification, priority floor protection, contradiction logging, and SafeSignal rule validation.
- `sarvamService.ts` : Cloud API client for Sarvam AI speech-to-text and chat completion.
- `lexiconData.ts` : Curated Hinglish lexicon categorized into physical violence, weapons, coded orders, confinement, and location stems.
- `types/index.ts` : Shared TypeScript models for reports, dispatches, telemetry, and triage state.

5. Tactical Fleet Dispatch & Geodesic Proximity Mathematics

Helpline responders are calculated dynamically relative to the caller's live GPS coordinates using the Great-Circle Haversine formula to compute exact ground distance and realistic urban arrival ETAs:

Geodesic Distance (Haversine Formula):

$$a = \sin^2(\Delta\varphi / 2) + \cos(\varphi_1) \cdot \cos(\varphi_2) \cdot \sin^2(\Delta\lambda / 2)$$

$$d = 2 \cdot R \cdot \operatorname{atan2}(\sqrt{a}, \sqrt{(1 - a)}) \quad [\text{where } R = 6{,}371 \text{ km earth radius}]$$

Dynamic Traffic-Weighted Urban ETA:

$$\text{ETA (minutes)} = \text{round}((d / v_{\text{responder}}) \times 60 \times T_{\text{traffic}} + T_{\text{turnout}})$$

- 112 Police PCR: $v = 32$ km/h, $T_{\text{turnout}} = 1.5$ min
- 108 ALS Ambulance: $v = 28$ km/h, $T_{\text{turnout}} = 2.0$ min
- 181 Sakhi Shelter: $v = 25$ km/h, $T_{\text{turnout}} = 3.0$ min
- DLSA Legal Aid: $v = 22$ km/h, $T_{\text{turnout}} = 4.0$ min

AGENCY / SERVICE	DIAL CODE	DEFAULT STATION / UNIT	TARGET DISPATCH PROTOCOL
Police Emergency (PCR)	`112`	PCR Van #DL-112-4	Immediate rapid interception; siren suppression if caller is in active domestic confinement.
Medical Ambulance (ALS)	`108`	Trauma Response Unit #108-B	Staged 200m away until police secure perimeter if weapons or active assault are flagged.
Sakhi One-Stop Center	`181`	OSC Crisis Intervention Van	Safe temporary shelter, trauma counselor, protective escort, medical examination referral.
Legal Services Authority	`15100`	DLSA Duty Advocate	Section 12 PWDVA protection orders, free legal aid, interim ex-parte maintenance.

6. Dynamic Situational Awareness & Unknowns Engine

In high-stress triage, CAD consoles must not present static question lists. SAATHI maintains a real-time situational checklist that dynamically verifies key operational unknown vectors based on speech input:

SITUATIONAL UNKNOWN VECTOR	VERIFICATION CONDITION	STATUS TRANSITION
Precise Address / Location Lock	GPS lock received or street/colony/gali matched in transcript.	`? Unconfirmed` → `✓ Verified Location`
Weapons or Implements Present	Weapons detected (`chaku` , `talwar` , `bandook`) or verified absent via safe inquiry.	`? Unconfirmed` → `⚠ Weapon Confirmed` / `✓ Cleared`
Active Physical Violence Now	Assault tokens detected (`maar raha hai` , `khoon` , `toda`) vs past report.	`? Unconfirmed` → `🚨 In Progress (P1)`
Abuser in Same Room / Coded Call	Food code, whispers, or explicit prompt indicating abuser proximity.	`? Unconfirmed` → `⚠ Danger: In Room (Yes/No Only)`
Medical Distress & Bleeding	Injury cues matched (`behosh` , `dard` , `chot`).	`? Unconfirmed` → `🚑 Medical Dispatch Req`
Children / Minors on Premises	Presence of infants/children confirmed through direct safe inquiry.	`? Unconfirmed` → `✓ Children Verified`

7. The SafeSignal Protocol & Direct Operator Inquiries

When a caller is in distress, standard questioning protocols often provoke further violence. SafeSignal enforces 10 strict invariants across all operator and automated prompts:

The 10 Inviolable Safety Rules (R1–R10)

1. **R1 (Binary Answers):** Strictly Yes/No only. No open questions.
2. **R2 (Brevity):** Question length must be ≤ 14 words.
3. **R3 (No Abuser Names):** Never say "pati", "husband", or abuser names.
4. **R4 (No Weapon Mentions):** Never ask "kya unke paas chaku hai".
5. **R5 (No Movement Demands):** Never command caller to run or open doors.
6. **R6 (No Volume Commands):** Never ask the caller to speak louder.
7. **R7 (No Dangerous Callbacks):** Never schedule an unannounced callback.
8. **R8 (Atomic Delivery):** Strictly one question at a time.
9. **R9 (Zero PII Ingestion):** No asking for Aadhaar, bank details, or IDs.
10. **R10 (No Promises):** Never promise police will arrive in 2 minutes.

Direct Operator Inquiry Channel

Operators can type custom questions or select verified presets. Once sent, the question renders directly on the citizen's mobile interface with high-contrast 1-touch response buttons:

USER MOBILE DISPLAY PREVIEW:

💬 **Operator: "Kya darwaza band hai?"**

[HAAN / YES] [NAHI / NO]

8. Core Data Schemas & TypeScript Interfaces

```
export interface EmergencyReport {
  id: string;
  source: 'voice_call' | 'silent_chat' | 'ivr_handoff';
  caller: { id: string; maskedPhone: string; language: string; location?: LocationTelemetry; };
  transcript: TranscriptTurn[];
  triage: StructuredTriage;
  dispatchState: AgencyDispatchState;
  contradictions: ContradictionRecord[];
  safeQuestions: SafeSignalQuestion[];
  timestamps: { initiated: number; locked?: number; closed?: number; };
}

export interface StructuredTriage {
  priority: 'P1_CRITICAL' | 'P2_URGENT' | 'P3_STANDARD' | 'P4_DEFERRED';
  category: 'domestic_violence' | 'physical_assault' | 'stalking_harassment' | 'sexual_assault' | 'coded_distress
  threatLevel: 'extreme' | 'high' | 'moderate' | 'low';
  flags: {
    codedCallDetected: boolean;
    urgencyFloorLocked: boolean;
    activeAssault: boolean;
    weaponsIdentified: boolean;
    childrenInDanger: boolean;
  };
  confidence: number;
  extractedEntities: {
    locationPhrase?: string;
    landmark?: string;
    weaponTypes?: string[];
    threats?: string[];
  };
}
```

9. Verification & Acceptance Test Coverage

The codebase is verified against **21 rigorous automated acceptance tests** covering every architectural requirement and safety constraint specified by the national helpline mandate:

SUITE NAME	TESTS	KEY COVERAGE AREAS	STATUS
Coded-Call Suite	5	Food delivery orders, wrong numbers, whisper cues, safe yes/no constraints.	PASSED (5/5)
Urgency Floors Suite	4	Monotonic priority locking, denial under duress resistance, weapon triggers.	PASSED (4/4)
SafeSignal Rules Suite	6	R1–R10 invariant checking, word count limits, PII filters, no-promise enforcement.	PASSED (6/6)
Location & Lexicon Suite	4	Colony/landmark parsing, pre-token verb trimming, Hinglish normalization.	PASSED (4/4)
Operator CAD & Multi-Agency	2	Simultaneous 4-agency dispatch, live ETA calculations, handoff packet generation.	PASSED (2/2)

10. Operational Impact & Deployment Readiness

By reducing emergency triage latency from the national average of **3.5 minutes down to under 10 seconds**, SAATHI prevents vital information loss during dropped or intercepted calls. Its quote-grounded evidence links eliminate operator cognitive bias, while its deterministic safety invariants ensure that women in the most

acute danger—those forced to speak in whispers or codes—receive the fastest, safest, and most coordinated response possible across Indian emergency services.
